How to Write a High-Quality Research Report

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Min Song

IEEE Format

Google: IEEE Format

https://www.ieee.org/conferences/publishing/templates.html

Microsoft Word
US letter
A4

LaTeX Template Instructions
Template
LaTeX Bibliography Files

Overleaf

https://www.overleaf.com/gallery/tagged/ieee-official

Title

- Either too long or too short is not good
- Informative – highlight the major product using a few keywords
  - Message Coverage Maximization in Infrastructure-Based Urban Vehicular Networks
  - ROP: Near-Optimal Rendezvous for Dynamic Spectrum Access Networks
  - Analysis of the On-Demand Spectrum Access Architecture for CBRS Cognitive Radio Networks
- A good title helps people to find your paper in Google
- Author names, email, affiliation

Xiaochan Xue[1], Shucheng Yu[2], Min Song[3]
Department of Electrical and Computer Engineering, Stevens Institute of Technology, Hoboken, NJ 07030, United States
Email: [1]xxue2@stevens.edu, [2]syu19@stevens.edu, [3]msong6@stevens.edu

Abstract

A concise summary of the research problem, solution, unique features of the solution, and results. Typically, one paragraph, 200 - 250 words.

An on-demand spectrum access cognitive radio network offers spectrum services to users so that users can dynamically set up application-oriented virtual topologies to support user applications. In this paper, we develop a mathematical model for the on-demand spectrum access architecture for cognitive radio networks based on the citizens broadband radio service (CBRS). The model can be used to estimate the blocking probability of spectrum demands from priority access license users, the network capacity, and the number of free spectrum bands available for lower priority general authorized access users. The performance evaluation indicates that the results from the mathematical model match simulation results well, which validates the accuracy of our model.

5 – 6 keywords

Section I. Introduction

This section introduces the background, importance, and challenges of the research problem. For a short paper like a course project report, this section can also include the related work and your main contributions.

At the end of the section, you need to outline how the rest of the paper is organized. This will help the readers to understand the structure of the report. Section I could be 1-1.5 pages.

The remainder of the paper is organized as follows. Section II gives an overview of the ODSA architecture and service provisioning for ODSA. Section III presents the mathematical model for ODSA and Section IV presents performance evaluation in a randomly generated CBRS-based CRN. Section V concludes the paper.

Section II. Problem Statement

This section formally defines the research problem. In many papers, this section defines the system model and mathematically formulates the problem.

The language in this section should be concise, accurate, and clear.
Do not beat around the bush without a focus. It is not a continuation of Section I.
It should clearly define the problem you are solving in this paper. Yes, just one problem, not two problems. In many cases, this section also defines the system/network model.

This section could be 0.5 – 1 page.

Section III. Solution

This section presents your work. It could be a new model, a new algorithm, a new architecture, a new design, a new protocol, or a new system. The section title should reflect your work.

When presenting your work, you also need to elaborate on the rationale of the new design: why your model/algorithm/design/protocol/system can solve the problem better than the existing work.

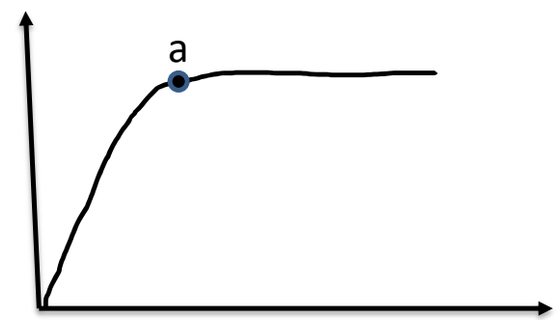
This is the most important section of the report. Please make sure you describe the novel aspects of your new model/algorithm/design/protocol/system clearly so that people can easily understand and appreciate it. This section could be 3-4 pages.

Section IV. Numerical Results and Analysis

This section provides numerical results to support your model/algorithm/design /protocol/system. To make your results more convincing, you need to compare your results with others or compare the analytical results with simulation results.

You may obtain the results through simulations, real-system experiments, or theoretical analysis. Very often, people use 5 – 7 figures and/or 1 – 3 tables to present numerical results. Each figure/table must be easy to follow and explained thoroughly.

Notice that you must provide the needed details on how you obtained the results. This section could be 2-3 pages.



Section V. Conclusion

This section summarizes the paper. Do not copy and paste the Abstract. This section could be just one big paragraph.

We have developed a mathematical model for a spectrum sharing architecture for cognitive radio networks, on-demand spectrum access (ODSA). We have applied the ODSA architecture and the developed model to citizens broadband radio service (CBRS) based cognitive radio networks, to estimate the blocking probability of user demands and network capacity. The performance evaluation indicates that the results from the mathematical model match simulation results well. We have also found that there are abundant free spectrum bands for lower priority general authorized access users in CBRS-based networks when priority access license users adopt the ODSA architecture and have moderate traffic loads.

References

At least 6 (individual project) or 12 (team project) references published in the last five years in IEEE/ACM journals or conferences.

All references must be cited in the paper body.

Do not copy and paste any single sentence/table/figure from any references or sources without citations.

Do not use any sentences generated from online tools or applications.

Acknowledgment

The research of XXXX's work is supported (mentored) by XXXX. However, any opinion, finding, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of XXXX.

Common Mistakes

- Long sentences -> hard to follow; Sloppy English
- Beat around the bush -> lack of a point
- There is no clear coherent connection between paragraphs/sections
  -> Lack of logic
- Ambiguous statements -> can be interpreted in different ways. Quantitative descriptions are always preferred
- Overstate contributions without justification or supporting evidence
- Inconsistent usage of terms or acronyms; Acronyms are not spelled out at the 1st time
- Numerical results lack a convincing justification